

# ENGINEERING BULLETIN

## Technical Information

### Various Chiller Monitoring and Data Logging Solutions

### Only Monitoring

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Dear Colleague,

This bulletin is to provide brief information regarding various Chiller Monitoring & Data Logging solutions Kirloskar chillers can offer to customers.

**KCPL Chiller's Monitoring and Logging solutions are as follows:**

1. Chiller HMI replication on PC
2. Chiller monitoring & data logging using SCADA
3. BMS Connectivity
4. Konnect
5. Chiller Trip Feedback.(Potential free contact)

#### **1. Chiller HMI replication on PC**

In this option, all pages of chiller controller screens are replicated as it is on PC through LAN network at site. It is real time monitoring of chiller HMI. The 13" or 15" (New Screen) HMI needs to be connected to existing LAN network at site or directly to assigned PC. Once configured HMI screens can be viewed on the PC using software provided by KCPL. Multiple chillers can also be monitored by assigning different IPs. Note following bullet points for this feature:

- This option is available only for TurboTEK & TurboLEV series which are with 13" or 15" (New Screen) HMI.
- It's a standard feature without any additional cost.
- Live monitoring of chiller parameters & alarms.
- Write access for User level set-points, like chilled water target.
- Data logging not possible with this option

#### **2. Chiller monitoring & data logging using SCADA**

This option is applicable for any chiller series (with or without 13" or 15" HMI). In this option, KCPL shall provide, Application developed in SCADA software along with its RUN license, which needs to install on assigned PC. Again, in this case also chiller connectivity shall be through Ethernet cable. Note following bullet points for this feature:

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### Monitoring, Data logging & report generation

Monitoring, Data logging & report generation can be done by third party BMS

Data logging, report generation & alarm intimation

- This option is available with all models.
- Live monitoring of chiller parameters & alarms.
- Write access for User level set-points, like chilled water target.
- **Data logging & report generation available.**
- Integration of additional devices with soft communication provision is feasible (e.g. Energy meter, BTU meter etc.)
- Customized programming on case to case basis required.
- The pricing need to be confirmed from engineering as it varies with number of chillers, additional devices, graphics development, etc.

### 3. BMS Connectivity

If separate BMS system is available and customer wants to monitor chillers using it, we have standard connectivity option as below:

- Modbus over RS485: This is by default available in all screw chillers, but needs to be specifically mentioned at the time of ordering in case of TurboTEK & TurboLEV chillers.
- Modbus over TCP/IP: If this connectivity option is required, needs to be specifically mentioned at the time of ordering in case of any series.(pco-web card need to consider at additional cost)
- BACnet: If this connectivity option is required, needs to be specifically mentioned at the time of ordering in case of any series.

### 4. Konnect

It is web based connectivity option by using Konnect device. Konnect communicates with chiller controller and sends the chiller data to cloud server using wireless mobile network connection. This is offered as a standard feature for all new chillers except KDC, KCM double & triple circuit models.

- Chiller parameters, user set points, alarms can be viewed by logging in through Konnect web portal.
- It is not exactly real time monitoring, parameters updates every 15 minutes but the real advantage is, it can be viewed from anywhere through internet.
- Daily log book mail sent automatically to customer pre-configured mail ID.

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### Trip Status

- Similarly Alarm alert mails also sent as & when alarm generates.

#### **5. Chiller Trip Feedback.**

- This option is available with all models.
- Only potential free contact provided.(Customers can connect hooter or lamp at customer end, if required optional Hooter will be provided at additional cost)

**Note: From options 2, 3 & 4 above only one can be offered at a time.**

With our continuous efforts for feature enhancement watch out for more bulletins regarding this in near future.